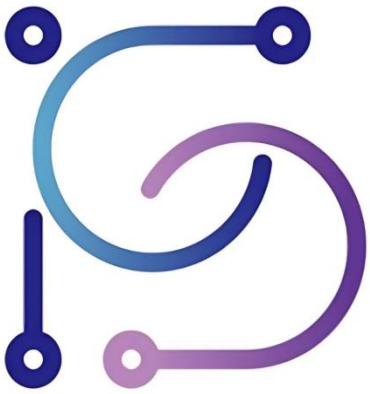


INDIESEMIC

ISC-nRF52840-A

An advanced multi-protocol ultra-low power BLE Module

DATASHEET

Version: V2.1

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Revision history:

Revision	Description	Date
1.0	– Initial Release	25/09/2023
2.0	– Restructured datasheet – Added FAQ – Power Consumption data – Reflow Diagram	20/04/2026
2.1	– Applied Copyright – Updated Pinout Diagram – Updated Footprint	13/05/2026

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1. Description

The ISC-nRF52840-A is a compact wireless module based on the Nordic nRF52840 multiprotocol SoC. It integrates a 64 MHz ARM® Cortex-M4 processor with Floating Point Unit (FPU), 1 MB Flash, and 256 kB RAM, providing sufficient resources for advanced wireless applications and protocol stacks. The module supports multiple radio standards, including Bluetooth® 5.4 Low Energy, Bluetooth Mesh, Thread, Zigbee, IEEE 802.15.4, ANT, and proprietary 2.4 GHz protocols. Its transceiver offers –95 dBm sensitivity (1 Mbps BLE), –103 dBm in long-range mode, and programmable transmit power up to +8 dBm, enabling reliable long-range wireless communication.

The module includes a rich set of digital and analog interfaces such as USB 2.0 with 12 Mbps Speed, QSPI (32 MHz), SPI, I^2C, I^2S, PDM, PWM, ADC, QDEC, RTC, and up to 46 configurable GPIOs. It also supports an NFC-A tag interface for simplified pairing and commissioning. Security features are implemented through ARM TrustZone® CryptoCell-310 and hardware accelerators for AES and other cryptographic algorithms.

Optimized for low-power operation, the ISC-nRF52840-A operates from 1V8 to 3V3, with ultra-low current consumption making it suitable for battery-operated devices such as wearables, sensor networks, smart home products, and industrial IoT applications requiring secure and energy-efficient wireless connectivity.

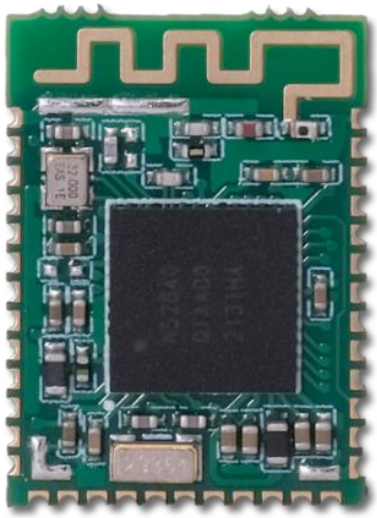


Figure 1- ISC-nRF52840-A

2. Features

- Bluetooth® 5, IEEE 802.15.4-2006, 2.4 GHz transceiver
  - -95 dBm sensitivity in 1 Mbps Bluetooth low energy mode
  - -103 dBm sensitivity in 125 kbps Bluetooth low energy mode (long range)
  - -20 to +8 dBm TX power, configurable in 4 dB steps
  - Supported data rates:
    - Bluetooth 5 – 2 Mbps, 1 Mbps, 500 kbps, and 125 kbps
    - IEEE 802.15.4-2006 – 250 kbps
    - Proprietary 2.4 GHz – 2 Mbps, 1 Mbps
  - RSSI (1 dB resolution)
- Arm® Cortex ® -M4 32-bit processor with FPU, 64 MHz
  - 212 EEMBC CoreMark ® score running from flash memory
  - 52 µA/MHz running CoreMark from flash memory
  - Watchpoint and trace debug modules (DWT, ETM, and ITM)
  - Serial wire debug (SWD)
- Flexible power management
  - 1.8 V to 5.5 V supply voltage range(Upto 3.3V for VDD and upto 5.5V For VBUS)
  - On-chip DC/DC and LDO regulators with automated low current modes
  - 1.8 V to 3.3 V regulated supply for external components
  - Automated peripheral power management
  - Fast wake-up using 64 MHz internal oscillator
- 1 MB flash and 256 kB RAM
- Advanced on-chip interfaces
  - USB 2.0 full speed (12 Mbps) controller
  - QSPI 32 MHz interface
  - High-speed 32 MHz SPI
  - Programmable peripheral interconnect (PPI)
  - 46 general purpose I/O pins
  - EasyDMA automated data transfer between memory and peripherals
  - Type 2 near field communication (NFC-A) tag with wake-on field
- 128- bit General purpose AES engine (encrypt/decrypt, sign/verify)
- Supports ECB, CBC, CMAC/CBC-MAC, CTR, CCM/CCM Encryption
- 12-bit, 200 ksps ADC – 8 configurable channels with programmable gain
- 64 level comparators
- Four 4 channel pulse width modulator (PWM) units with EasyDMA
- Audio peripherals – I2S, digital microphone interface (PDM)
- Five 32-bit timers with counter mode
- Up to four SPI masters/three SPI slaves with EasyDMA
- Up to two I2C compatible two-wire master/slave
- Two UART (CTS/RTS) with EasyDMA
- Quadrature decoder (QDEC)
- Three real-time counters (RTC)

3. Applications

- Advanced computer peripherals and I/O devices
  - Mouse
  - Keyboard
  - Multi-touch trackpad

- Internet of things (IOT)
  - Smart home products
  - Industrial mesh networks
  - Smart city infrastructure

- Advanced wearable devices
  - Connected watches
  - Advanced personal fitness devices
  - Wearables with wireless payment
  - Connected health
  - Virtual/Augmented reality applications

- Interactive entertainment devices
  - Advanced remote controls
  - Gaming controller

- Personal area networks
  - Health/Fitness sensor and monitor device
  - Medical device

4. Block Diagram

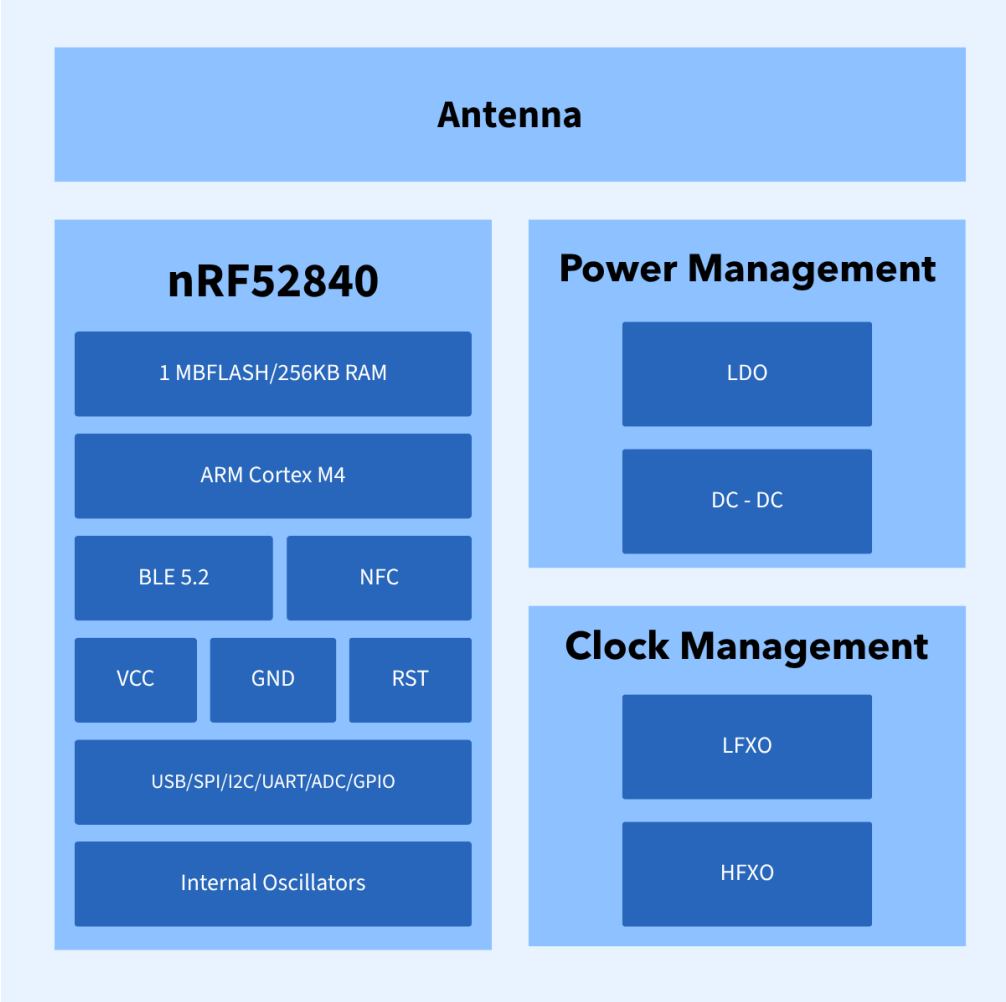


Figure 2- Block Diagram

5. Module Pin-out

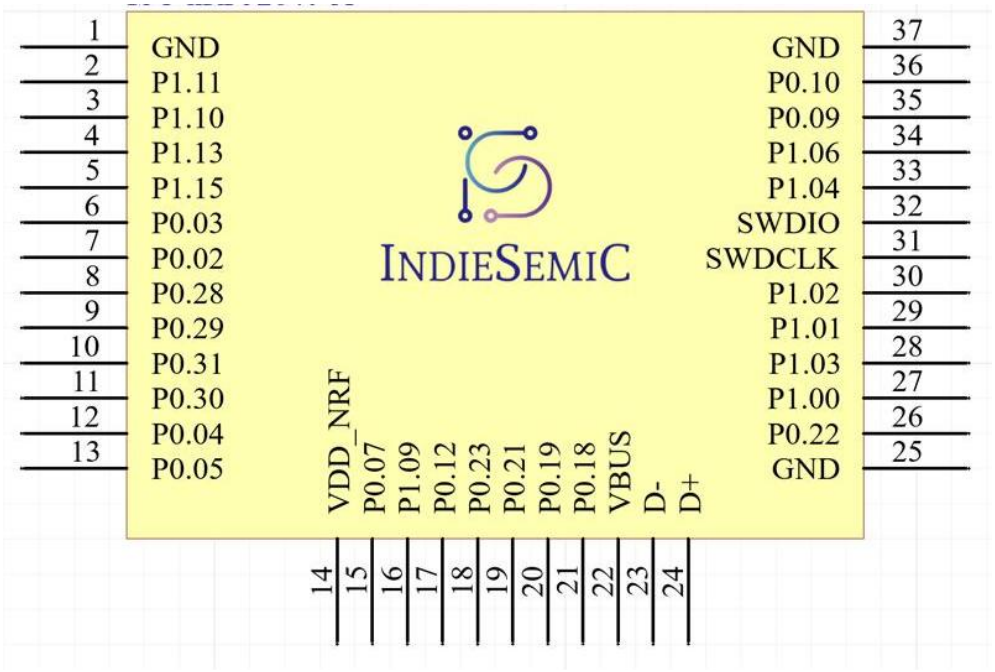


Figure 3- Module Pinout (back view-I)

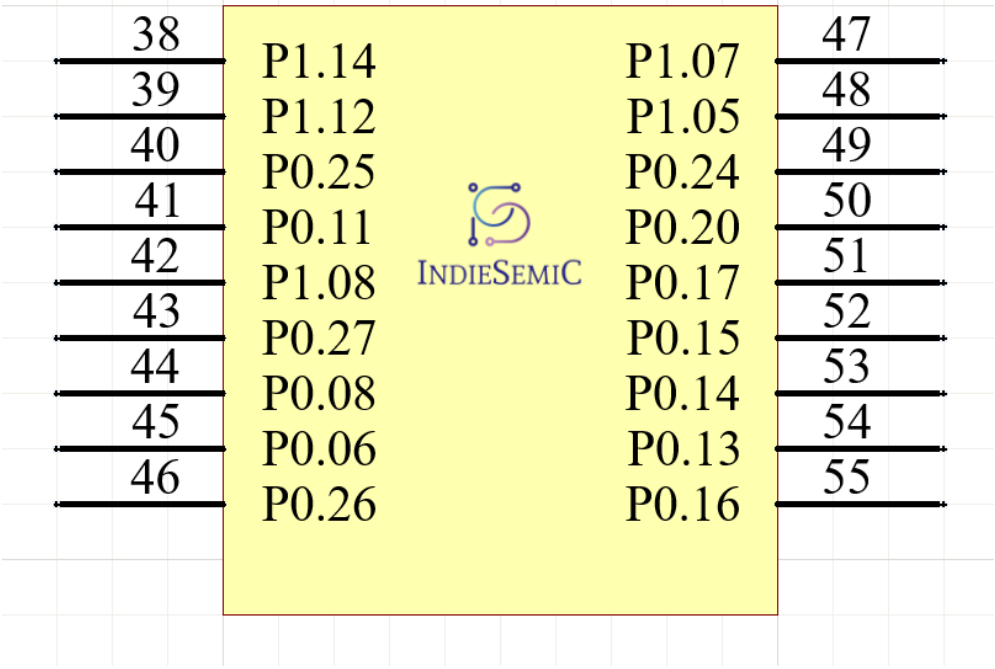


Figure 4- Module Pinout (back view-II)

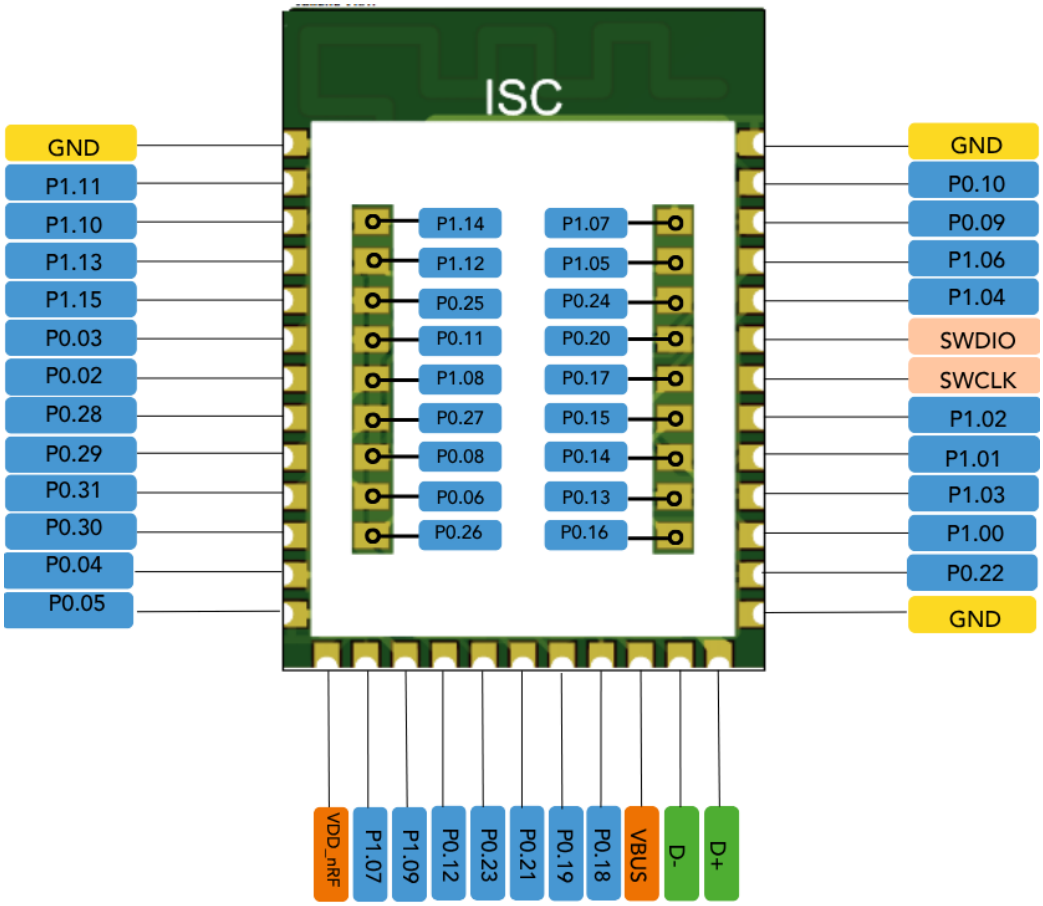


Figure 5- Pin Diagram (back view)

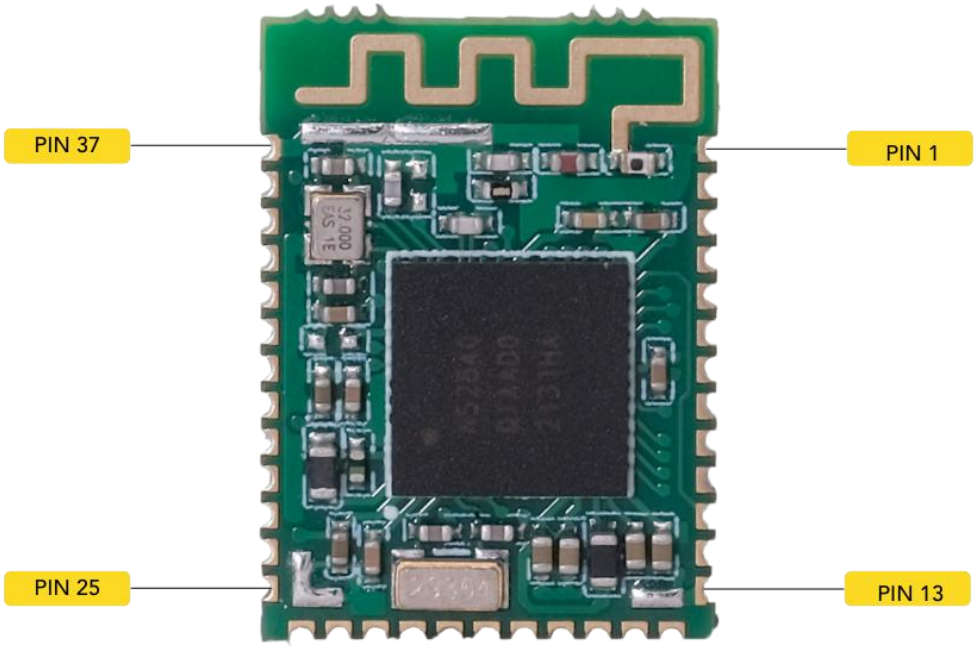


Figure 6- Pin Diagram (front view)

6. PCB Footprint

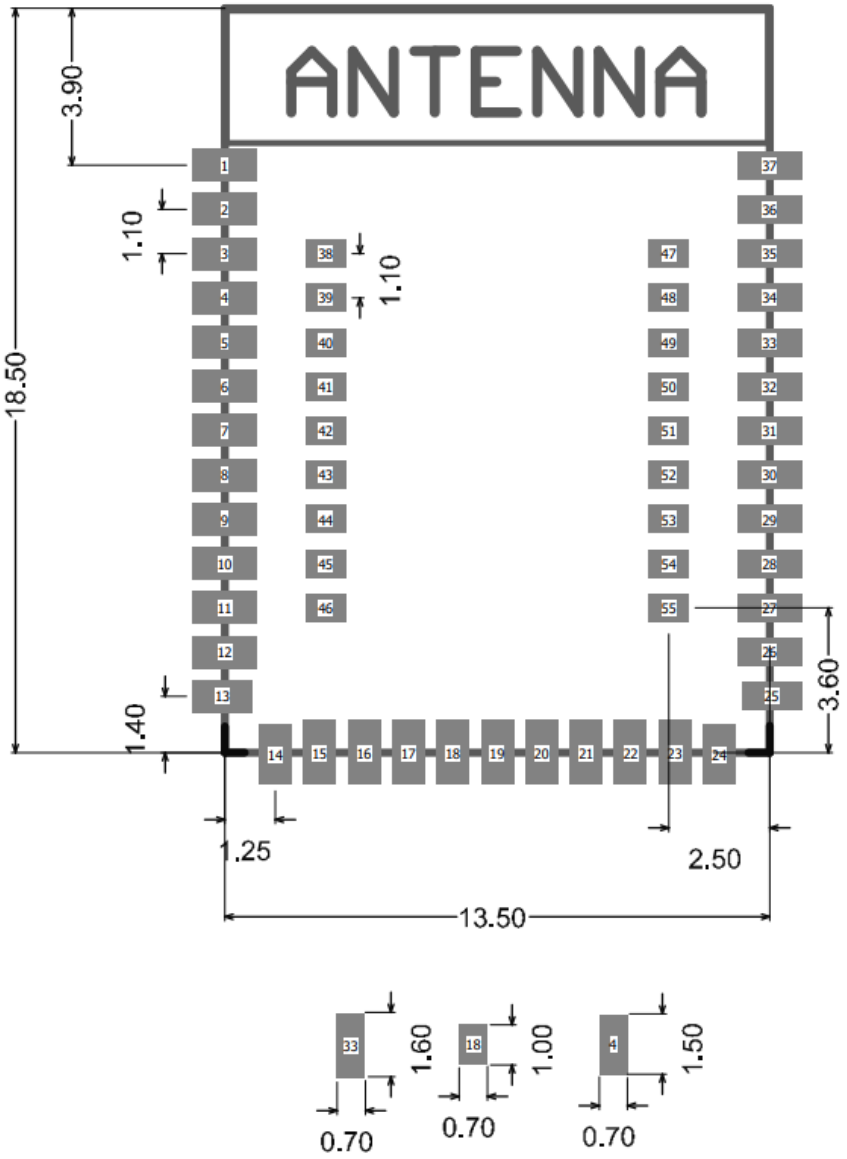


Figure 7- PCB Footprint (back)

7. Pin Descriptions

Pin No.	Name	I/O Type	Description
1	GND	Power	Ground
2	P1.11	Digital I/O	General-Purpose I/O
3	P1.10	Digital I/O	General-Purpose I/O
4	P1.13	Digital I/O	General-Purpose I/O
5	P1.15	Digital I/O	General-Purpose I/O
6	P0.03 / AIN1	Digital I/O / Analog Input	GPIO Analog Input
7	P0.02 / AIN0	Digital I/O / Analog Input	GPIO Analog Input
8	P0.28 / AIN4	Digital I/O / Analog Input	GPIO Analog Input
9	P0.29 / AIN5	Digital I/O / Analog Input	GPIO Analog Input
10	P0.31 / AIN7	Digital I/O / Analog Input	GPIO Analog Input
11	P0.30 / AIN6	Digital I/O / Analog Input	GPIO Analog Input
12	P0.04 / AIN2	Digital I/O / Analog Input	GPIO Analog Input
13	P0.05 / AIN3	Digital I/O / Analog Input	GPIO Analog Input
14	VDD_NRF	Power	Power Supply
15	P0.07 / TRACECLK	Digital I/O	General-Purpose I/O Trace Buffer Clock
16	P1.09 / TRACE DATA	Digital I/O / TRACE DATA 3	General-Purpose I/O Trace Buffer Trace Data[3]
17	P0.12 / TRACE DATA	Digital I/O / TRACE DATA 1	General-Purpose I/O Trace Buffer Trace Data [1]
18	P0.23	Digital I/O	General-Purpose I/O
19	P0.21	Digital I/O	General-Purpose I/O
20	P0.19	Digital I/O	General-Purpose I/O
21	P0.18 / RESET	Digital I/O	General-Purpose I/O / System RESET
22	VBUS	Power	USB Reference Voltage
23	D-	Digital I/O	USB D-
24	D+	Digital I/O	USB D+
25	GND	Power	Ground
26	P0.22	Digital I/O	General-Purpose I/O

27	P1.00 / TRACEDATA0	Digital I/O	General-Purpose I/O Trace Buffer Trace Data [0]
28	P1.03	Digital I/O	General-Purpose I/O
29	P1.01	Digital I/O	General-Purpose I/O
30	P1.02	Digital I/O	General-Purpose I/O
31	SWDCLK	Debug	Debug serial clock input
32	SWDIO	Debug	Debug data input/output
33	P1.04	Digital I/O	General-Purpose I/O
34	P1.06	Digital I/O	General-Purpose I/O
35	P0.09 / NFC1	Digital I/O	General-Purpose I/O / NFC Antenna Connection
36	P0.10 / NFC2	Digital I/O	General-Purpose I/O / NFC Antenna Connection
37	GND	Power	Ground
38	P1.14	Digital I/O	General-Purpose I/O
39	P1.12	Digital I/O	General-Purpose I/O
40	P0.25	Digital I/O	General-Purpose I/O
41	P0.11 / TRACEDATA2	Digital I/O	General-Purpose I/O / Trace Buffer Trace Data[2]
42	P1.08	Digital I/O	General-Purpose I/O
43	P0.27	Digital I/O	General-Purpose I/O
44	P0.08	Digital I/O	General-Purpose I/O
45	P0.06	Digital I/O	General-Purpose I/O
46	P0.26	Digital I/O	General-Purpose I/O
47	P1.07	Digital I/O	General-Purpose I/O
48	P1.05	Digital I/O	General-Purpose I/O
49	P0.24	Digital I/O	General-Purpose I/O
50	P0.20	Digital I/O	General-Purpose I/O
51	P0.17	Digital I/O	General-Purpose I/O
52	P0.15	Digital I/O	General-Purpose I/O
53	P0.14	Digital I/O	General-Purpose I/O
54	P0.13	Digital I/O	General-Purpose I/O
55	P0.16	Digital I/O	General-Purpose I/O

Table 1- Pin Descriptions

8. Electrical Specifications

8.1 Absolute Maximum Ratings

Parameter	Condition	Min.	Typical	Max.
Storage Temperature		-40		125 °C
ESD Protection	VESD			2000 V
Supply Voltage	VCC, VBus	-0.3 V		3.9 V
Voltage on Any I/O Pin		-0.3 V		3.63 V

Table 2- Maximum Rating

8.2 Recommended Operating Ratings

Parameter	Symbol	Min.	Typical	Max.	Unit
Operating Temperature	TA	-40	25	85	°C
Power Supply	VCC	1.7	3.3	3.6	V
Input Low Voltage	VIL	0		0.3×VCC	V
Input High Voltage	VIH	0.7×VCC		VCC	V

Table 3- Operating Ratings

9. Function Description

9.1.1 GPIO

The general purpose I/O is organized as one port with up to 46 I/Os enabling access and control of up to 46 pins through one port. Each GPIO can be accessed individually with the following user configurable features:

- Input/output direction
- Output drive strength Internal pull-up and pull-down resistors
- Wake-up from high- or low-level triggers on all pins
- Trigger interrupt on all pins
- All pins can be used by the PPI task/event system; the maximum number of pins that can be interfaced through the PPI at the same time is limited by the number of GPIOTE channels
- All pins can be individually configured to carry serial interface or quadrature demodulator signals
- All pins can be configured as PWM
- There are 6 ADC/LPCOMP input in the 46 I/Os

9.1.2 Two Wire Interface

The ISC-nRF52840-A can communicate with a bi-directional wired-AND bus with two lines (SCL, SDA). The protocol makes it possible to interconnect up to 127 individually addressable devices. The interface is capable of clock stretching and supporting data rates of 100 kbps, 250kbps and 400 kbps.

9.1.3 Flash Program I/Os

The module has two programmer pins, respectively SWDCLK pin and SWDIO pin. The two pin Serial Wire Debug (SWD) interface provided as a part of the Debug Access Port (DAP) offers a flexible and powerful mechanism for non-intrusive debugging of program code. Breakpoints and single stepping are part of this support.

9.1.4 Serial Peripheral Interface

The SPI interfaces enable full duplex synchronous communication between devices. They support a three-wire (SCK, MISO, MOSI) bi-directional bus with fast data transfers. The SPI Master can communicate with multiple slaves using individual chip select signals for each of the slave devices attached to a bus.

Control of chip select signals is left to the application through use of GPIO signals.

SPI Master has double buffered I/O data. The SPI Slave includes EasyDMA for data transfer directly to and from RAM allowing Slave data transfers to occur while the CPU is IDLE. The GPIOs are used for each SPI interface line and can be chosen from any GPIOs on the device and configured independently. This enables great flexibility in device pinout and efficient use of printed circuit board space and signal routing.

The SPI peripheral supports SPI mode 0, 1, 2, and 3. The module has 3 SPI ports
and their properties are as below:

Instance	Master/Slave
SPI0	Master
SPI1	Master
SPIS1	Slave

Table 4- SPI

9.1.5 UART's

The Universal Asynchronous Receiver/Transmitter offers fast, full-duplex, asynchronous serial communication with built-in flow control (CTS, RTS), support in hardware up to 1 Mbps baud. Parity checking is supported. Support

**__Note:__ GPIOs for SPI, TWI (I2C), and UART can be mapped to almost any available GPIO pins using flexible pin configuration.*

However, peripherals are grouped into shared instances (e.g., SPIM0/TWIM0/UARTE0 share the same hardware block). Only one function can be used per instance at a time. Therefore, SPI0, UART0, and I2C0 cannot be used simultaneously, but different instances (e.g., SPI0 + UART1 + I2C1) can be used together.

9.1.6 Analogue to Digital Converter

The 12-bit incremental Analogue to Digital Converter (ADC) enables sampling of
up to 8 external signals through a front-end multiplexer. The ADC has configurable input and reference pre-scaling, and sample resolution (8,10, and 12 bit).

*__*Note:__ The ADC module uses the same analogue inputs as the LPCOMP module. Only one of the modules can be enabled at a time.*

Module PIN Number	ISC-nRF52840-A PIN Number	Description
7	P0.02/AIN0	General Purpose I/O SAADC/COMP/LPCOMP Input
6	P0.03/AIN1	General Purpose I/O SAADC/COMP/LPCOMP Input
12	P0.04/AIN2	General Purpose I/O SAADC/COMP/LPCOMP Input
13	P0.05/AIN3	General Purpose I/O SAADC/COMP/LPCOMP Input
8	P0.28/AIN4	General Purpose I/O SAADC/COMP/LPCOMP Input
9	P0.29/AIN5	General Purpose I/O SAADC/COMP/LPCOMP Input
11	P0.30/AIN6	General Purpose I/O SAADC/COMP/LPCOMP Input
10	P0.31/AIN7	General Purpose I/O SAADC/COMP/LPCOMP Input

Table 5- ADC

9.1.7 Low Power Comparator (LPCOMP)

In System ON, the block can generate separate events on rising and falling edges of a signal or sample the current state of the pin as being above or below the threshold. The block can be configured to use any of the analogue inputs on the device. Additionally, the low power comparator can be used as an analogue wakeup source from System OFF or System ON. The comparator threshold can be programmed to a range of fractions of the supply voltage.

9.1.8 NFC

The NFC peripheral (referred to as the 'NFC peripheral' from now on) supports communication signal interface type A and 106 kbps bit rate from the NFC Forum.

Please Refer here for More Information related to protocol.

10. Radio Peripherals

The ISC-nRF52840-A integrates a highly flexible 2.4 GHz transceiver based on Nordic's nRF52840 SoC. It supports Bluetooth Low Energy (Bluetooth 5.4), Bluetooth Mesh, IEEE 802.15.4 (Thread/Zigbee), and proprietary 2.4 GHz protocols. The radio includes an on-chip balun with a single-ended 50 Ω antenna interface, enabling direct connection to the module's PCB antenna. Programmable transmit power up to +8 dBm and receiver sensitivity down to −103 dBm (long-range mode) allow robust wireless links while maintaining low current consumption.

10.1 Bluetooth Low Energy:

The ISC-nRF52840-A integrates full Bluetooth Low Energy (BLE) 5.x capability, enabling both high-throughput and long-range operation
Please Refer Nordic Doc's for More Information related to protocol.

10.2 Bluetooth Mesh:

The ISC-nRF52840-A fully supports Bluetooth Mesh networking on top of Bluetooth Low Energy.
Please Refer here for More Information related to protocol.

10.3 Thread:

The ISC-nRF52840-A supports Thread networking based on the IEEE 802.15.4 standard at 2.4 GHz.
Please Refer here for More Information related to protocol.

10.4 Matter:

The ISC-nRF52840-A is fully capable of running Matter, the unified smart home connectivity standard built on top of IPv6 and Thread.
Please Refer here for More Information related to protocol.

10.5 Zigbee

The ISC-nRF52840-A supports Zigbee networking using its IEEE 802.15.4 radio.
Please Refer here for More Information related to protocol.

10.6 ANT

The ISC-nRF52840-A supports the ANT wireless protocol in addition to Bluetooth Low Energy.
Please Refer here for More Information related to protocol.

11. Power Consumption Analysis:

The ISC-nRF52840-A module is engineered for superior energy efficiency in wireless environments. Performance metrics indicate that while the hardware is optimized for ultra-low power consumption, actual current draw is contingent upon specific firmware logic, radio transceiver states, and operational duty cycles. The profiles detailed below represent empirical measurements obtained during Bluetooth Low Energy (BLE) transmissions and system idle states.

11.1 Low Power / Idle Current Profile

In low-power conditions (no active radio transmission), the module demonstrates near datasheet-level efficiency.

Observed Characteristics:

- Peak Current: ~12 µA
- Average Current: ~0.58 µA
- Charge Consumption (10 s window): ~5.76 µC

Interpretation:

- The system remains mostly in low-power sleep states.
- Small periodic spikes are due to RTC wakeups or minimal background tasks.
- This aligns closely with System ON low-power expectations.



Figure 8- Low power / ideal current profile

11.2 BLE Advertising Current Profile

During BLE advertising, the module exhibits periodic current spikes corresponding to radio transmission events.

Observed Characteristics:

- Peak Current @4dBm: ~12 mA
- Average Current: ~586 µA
- Advertising Interval: ~100 ms
- Charge Consumption (10 s window): ~5.87 mC

Interpretation:

- The sharp spikes (~10–12 mA) represent TX bursts.
- The baseline (~1–2 µA to hundreds of µA depending on state) represents sleep/System ON idle between events.
- Average current is significantly influenced by advertising intervals
- shorter intervals drastically increase power consumption.

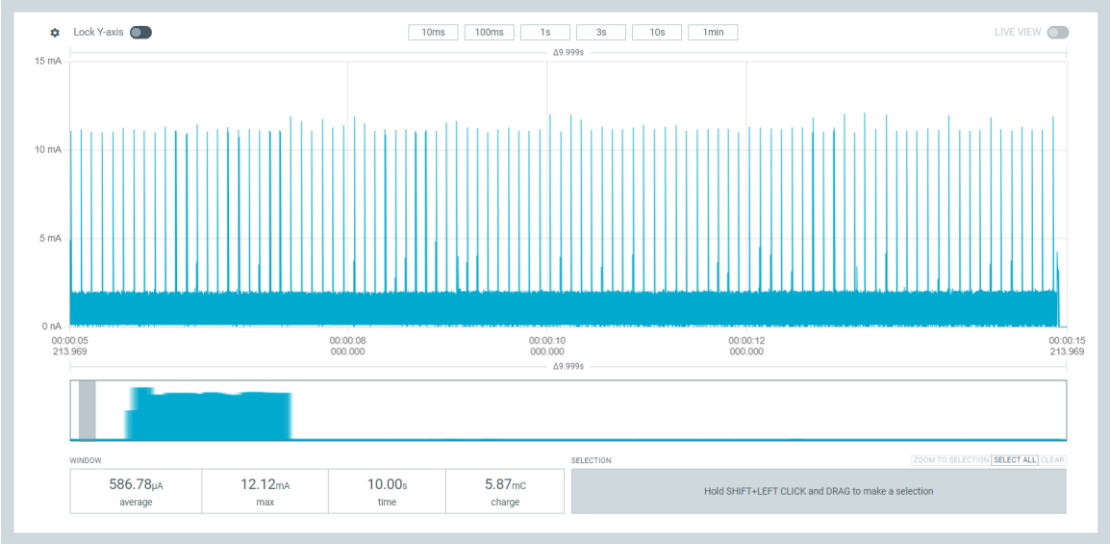


Figure 9- BLE profile

11.3 Summary

Mode	Peak Current	Average Current	Notes
BLE Advertising@4dBm	~12 mA	~586 µA	Dominated by TX bursts
Low Power (Idle)	~12 µA	~0.58 µA	Near System ON low-power
System OFF (Expected)	~0.4 µA	~0.4 µA	No RAM retention

Table 6- Power Consumption

12. Reflow Diagram:

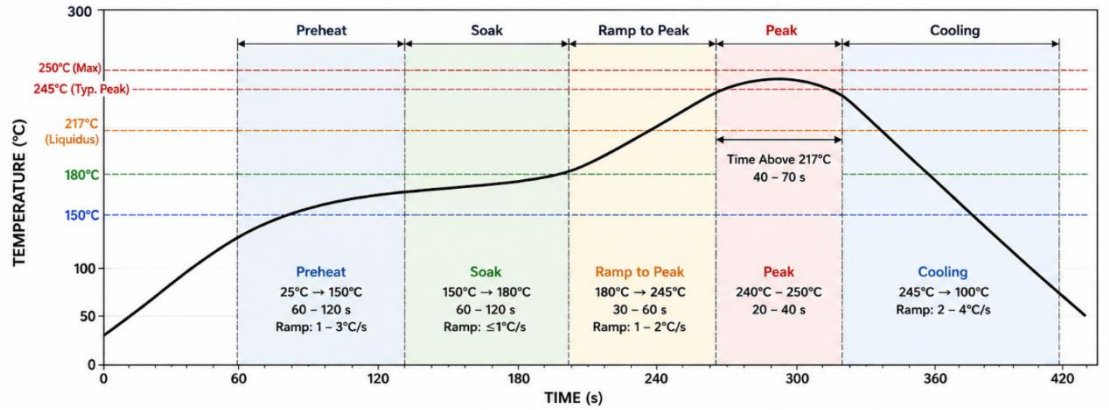


Figure 10 - Reflow Diagram

Stage	Temperature Range	Time Duration	Ramp Rate
Preheat	25°C → 150°C	60–120 sec	1–3°C/sec
Soak	150°C → 180°C	60–120 sec	≤1°C/sec
Ramp to Peak	180°C → 245°C	30–60 sec	1–2°C/sec
Peak Temp	240°C → 250°C	20–40 sec	—
Cooling	245°C → 100°C	—	2–4°C/sec

Table 7 - Reflow Rate

13. FAQs:

- **How To Program ISC Series of beacons & Module:**

To program the IndieSemiC Range of Modules/Beacons, use the segger J-Link debugger. For development and testing purposes we have used NRF52840 DK for programming below is short guideline for it.

Make the following connections between the Nordic DK and the target module:

- **SWDIO → SWDIO**
- **SWCLK → SWCLK**
- **RESET → RESET**
- **VCC (3.3 V) → VCC**
- **GND → GND**
- **Short VDD_nRF and VTG (or SWDSL)** on the DK to ensure the target receives correct reference voltage.

After wiring, power the module through the DK and use nRF Connect for Desktop (Programmer app) **or the** nrfjprog **command-line tool to flash firmware via SWD.**

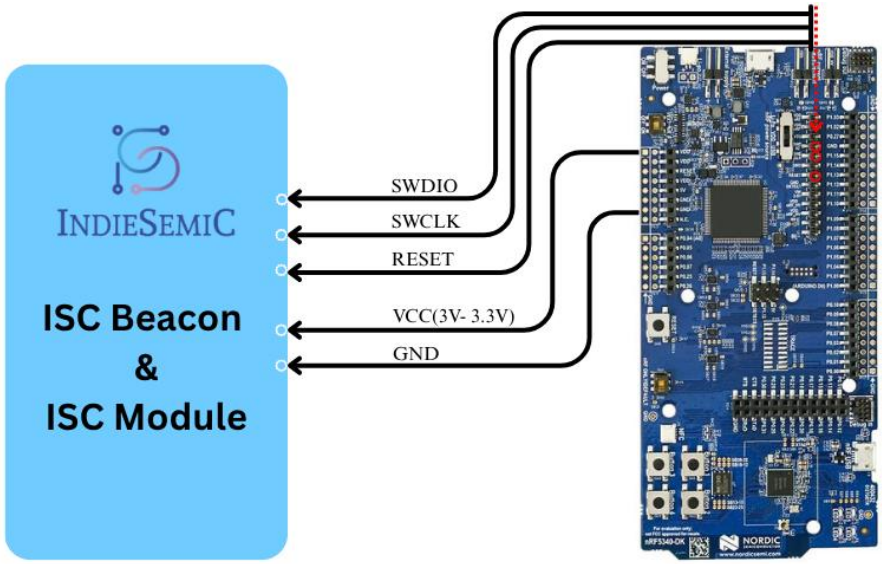


Figure 11- Programming ISC modules & beacons

- For a wider range of RF modules and evaluation kits, or to find an evaluation kit for this module:
  Evaluation Kit: EVK-ISC-nRF52840-A
  Other: RF Modules

- If you would like your schematic or Gerber files reviewed before Production, you can contact Nordic DevZone or reach out to us at **sales@indiesemic.com**.

Contact & Support:

For further technical assistance, customization requests, or queries regarding Indiesemic ISC Series Modules and Evaluation Kits, please contact our sales team:
Email: sales@indiesemic.com

We are committed to providing prompt responses and ensuring seamless integration of our solutions into your applications.

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